

1. The common - Ion Effect

举了 CH_3COOH 和 NaCH_3COO 的例子

Acetic acid Sodium acetate

有强有弱

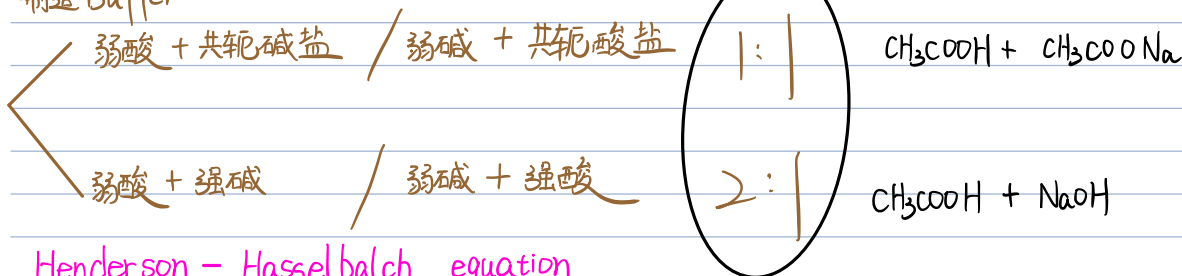
- “Whenever a **weak electrolyte** and a **strong electrolyte** containing a **common ion** are together in solution, the weak electrolyte **ionizes less than** it would if it were alone in solution.”

→ 计算不可忽略问题, 先假设可忽略, 算出来再比较

2. Buffers

- Solutions of a **weak conjugate acid-base pair** that resist **drastic changes** in pH are called **buffers**. 急剧的
- These solutions contain relatively **high concentrations** (10^{-3} M or more) of both the acid and base. Their concentrations are **approximately equal**.

制造 Buffer



Henderson - Hasselbalch equation

计算 Buffer PH

$$\text{pH} = \text{pK}_a + \log \left(\underset{\substack{\uparrow \\ \text{碱}}}{[\text{A}^-]} / \underset{\substack{\uparrow \\ \text{酸}}}{[\text{HA}]} \right)$$

两种相当于
也可以公式方程直接算

pH Range

- If one concentration is more than **10 times the other**, the buffering action is poor; this means that the pH range of a buffer is usually **$\pm 1 \text{ pH}$ unit from pK_a** .
between **$(\text{pK}_a - 1)$** and **$(\text{pK}_a + 1)$**

3. Titration 滴定

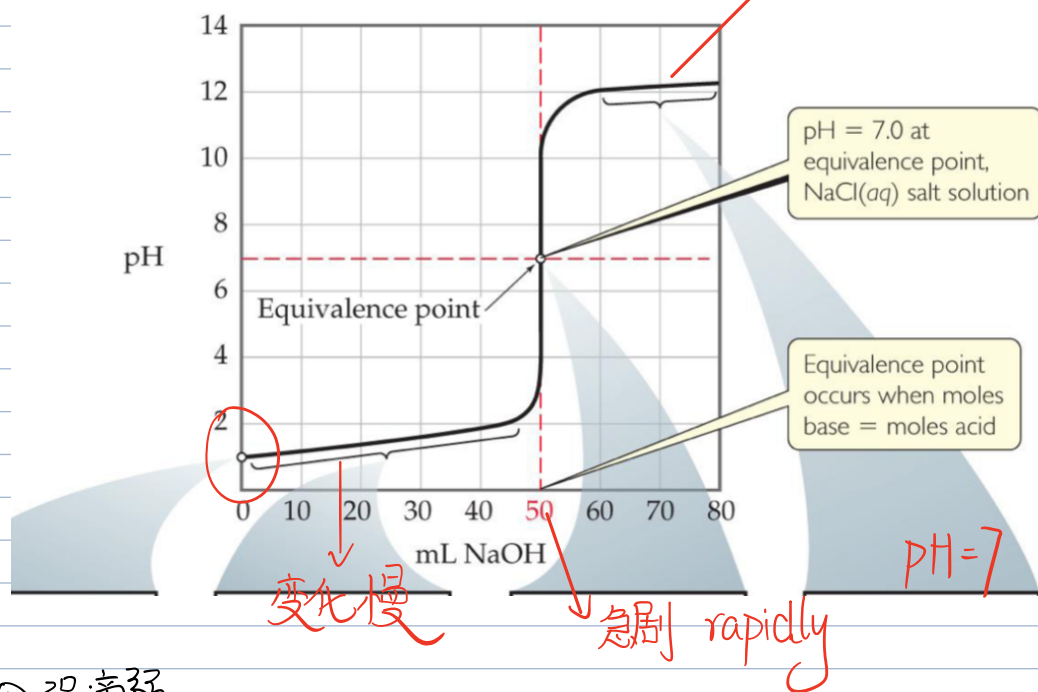
equivalence point: the amount of acid = base

酸滴碱 / 碱滴酸 (已知滴未知)

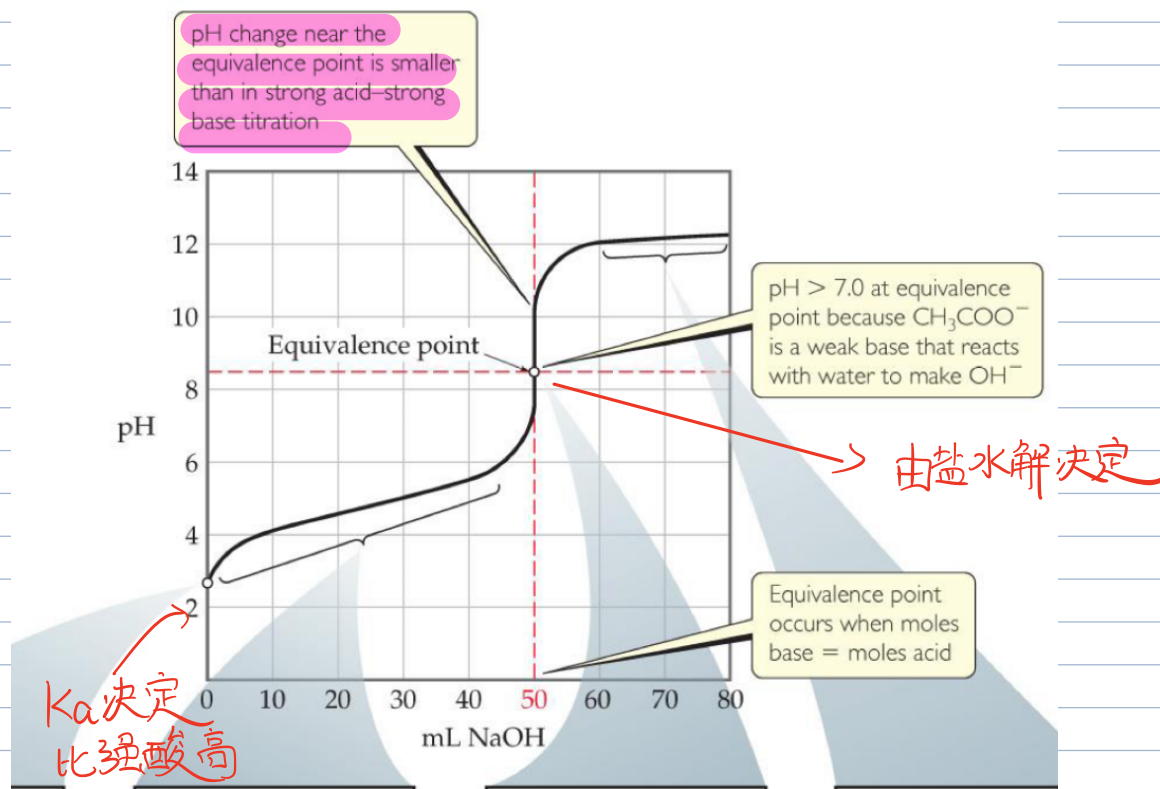
有几种类型

① 强滴强

equivalence point: 只有盐

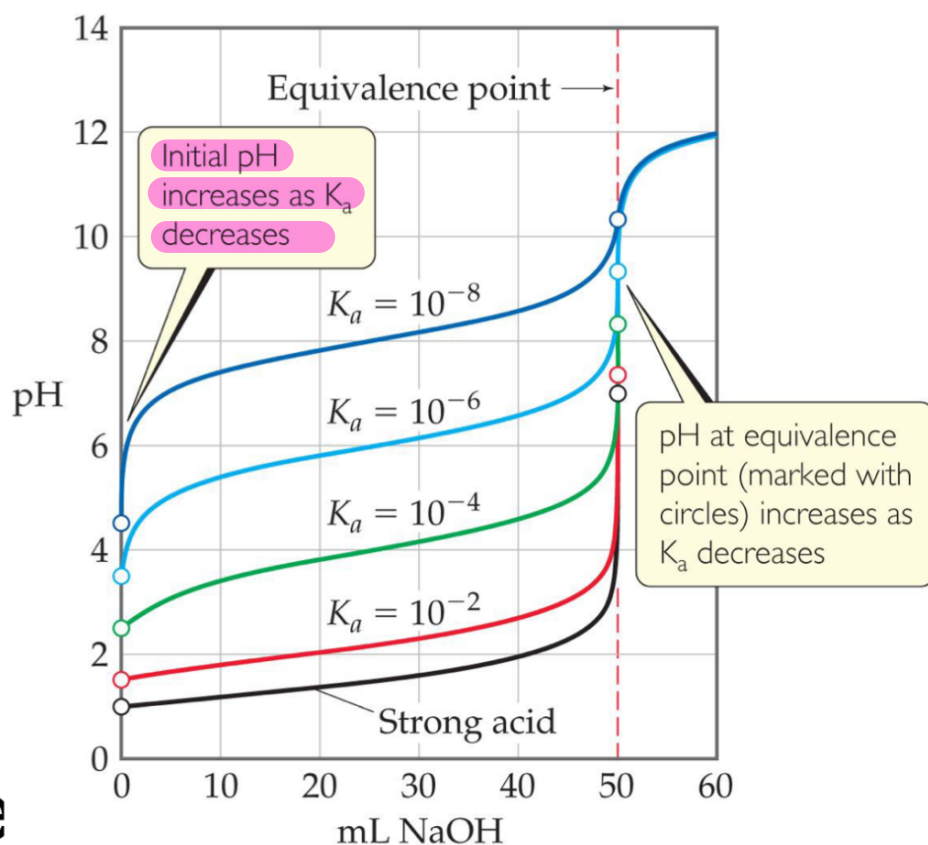


② 强滴弱



弱
↑
强

2



indicators

指示剂的选择

变色的pH范围包含

Equivalence point

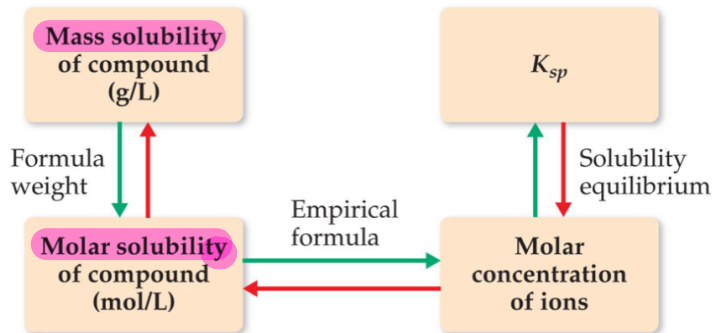
- For titration of weak acid by strong base, **phenolphthalein** is typically a better choice. 8.3 ~ 10
- For titration of weak base by strong acid, **methyl red** is typically a better choice. 4.2 ~ 6



Polyprotic acid 的滴定
多元酸
有多个这样的过程

4. K_{sp}

solubility — product constant



5. factors affect solubility

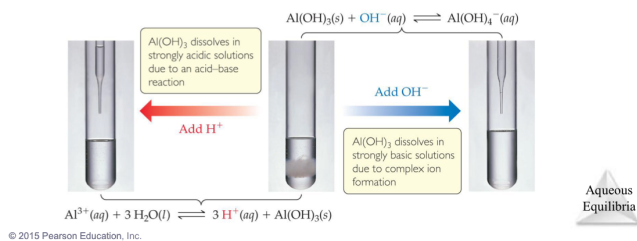
→ Common-Ion

→ PH

→ 形成 complex ion
与其中离子

Amphoteric

- **Amphoteric oxides and hydroxides** are soluble in strong acids or base, because they can act either as acids or bases.
- Examples are oxides and hydroxides of Al^{3+} , Cr^{3+} , Zn^{2+} , and Sn^{2+} .



- An **amphiprotic** substance can accept or donate a proton (H^+ ion).
- An **amphoteric** substance can behave as an acid and as a base.
- All **amphiprotic** substances are **amphoteric**, but **NOT** all **amphoteric** substances are **amphiprotic**.

判断沉淀 (precipitate) 是否形成 Q- K_{sp} 比较

选择指示剂的原则

equivalence point pH in range of

- 指示剂 PH
- or
- $pK_a \pm 1$
 - 指示剂 K_a